SUBSPACE-CONSTRAINED FEDERATED LEARNING WITH LOW-RANK ADAPTATION

Neranjan Senarath
Department of ECSE
Rensselaer Polytechnic Institute
Troy, NY 12180, USA
senarh@rpi.edu

Rohit Muralitharan
Department of Computer Science
Rensselaer Polytechnic Institute
Troy, NY 12180, USA
muralr@rpi.edu

Sadia Asif
Department of Computer Science
Rensselaer Polytechnic Institute
Troy, NY 12180, USA
asifs@rpi.edu

ABSTRACT

Federated low-rank adaptation methods are attractive for fine-tuning large models under communication and privacy constraints, but heterogeneous client data can induce geometric misalignment between local low-rank updates. We study whether this subspace misalignment leads to destructive aggregation and slower convergence in LoRA-based federated learning. We propose a subspace-regularized federated LoRA objective that encourages local client updates to remain close to a shared global reference subspace. We present a complete empirical evaluation on two pretrained models—RoBERTa-large and SmolLM-360M—over HellaSwag in a non-IID 10-client federated setting, across 3 random seeds (42, 43, 44), yielding 24 total experimental runs (4 methods $\times$ 3 seeds $\times$ 2 models). On RoBERTa-large, Subspace-Reg achieves the strongest mean best accuracy (0.454 ± 0.023), mean final accuracy (0.429 ± 0.011), and lowest final loss (1.363) across all three seeds, outperforming FedAvg, SVD redistribution, and FedSVD baselines by a large margin. On SmolLM-360M, FedAvg leads on accuracy, revealing that accuracy gains are model-dependent. Crucially, Subspace-Reg achieves near-perfect basis overlap (≈ 0.9999) on *both* models and across all seeds, versus 0.958–0.991 for all baselines, providing robust support for the geometric alignment hypothesis. The code is publicly available at GitHub repository.

1 INTRODUCTION

Federated Learning (FL) enables collaborative model training across decentralized clients without sharing raw data, thereby preserving privacy and reducing communication overhead (McMahan et al., 2017; Karimireddy et al., 2020). Among existing approaches, Federated Averaging (FedAvg) is the most commonly used optimization paradigm, relying on periodic averaging of local model updates. However, in realistic federated environments where client data distributions are heterogeneous, FedAvg is known to suffer from instability and slow convergence (Li et al., 2020). This statistical heterogeneity often leads to update drift across clients, which can degrade global model performance.

With the emergence of large foundation models, parameter-efficient fine-tuning techniques such as Low-Rank Adaptation (LoRA) (Hu et al., 2022) and its extension (Dettmers et al., 2023) have become increasingly popular in FL. LoRA constrains weight updates to a low-rank decomposition $\Delta W = BA$, significantly reducing communication and computational cost while preserving adaptation capacity. Recent works have explored integrating LoRA into FL frameworks to enable scalable fine-tuning of large language models (Bai et al., 2024; Lee et al., 2025; Chen et al., 2025). These

methods primarily focus on different aggregation strategies, including sum-product (SP), product-sum (PS), and SVD-based redistribution schemes.

In this project, we investigate a phenomenon that we hypothesize to be important but relatively underexplored: *subspace misalignment*. Under heterogeneous data distributions, different clients can learn low-rank updates that span distinct subspaces, which may reduce the effectiveness of naive aggregation when those updates are combined at the server (Lee et al., 2025; Bai et al., 2024). When aggregated naïvely, these updates may partially cancel each other, resulting in what we refer to as **silent cancellation**. Even when individual clients make substantial local progress, the aggregated global update may have reduced magnitude due to geometric misalignment between client subspaces; related observations about cancellation and reduced effective progress under federated heterogeneity have also been discussed in recent work (Mahla et al., 2024; Li et al., 2020). We hypothesize that this structural cancellation contributes to slow convergence in LoRA-based FL.

To address this issue, we propose a **Subspace-Regularized LoRA** framework. Our idea is to introduce a proximal alignment penalty that encourages each client's local low-rank basis to stay close to a shared global reference subspace. Specifically, we incorporate a Frobenius-norm proximal term that constrains local updates to remain close to the server's reference update, while optionally regularizing the basis matrices themselves. Intuitively, this encourages constructive aggregation rather than destructive interference, in the same spirit that FedProx uses proximal regularization to control client drift in standard federated optimization (Li et al., 2020).

Conceptually, our approach extends the idea of FedProx (Li et al., 2020), which controls update magnitude drift, to the geometric structure of low-rank subspaces. Instead of only restricting how far local parameters move, we attempt to regulate the *direction* of movement in the low-rank manifold. Through this project, we aim to both empirically and theoretically evaluate whether subspace alignment can improve convergence stability and aggregation efficiency in heterogeneous federated settings.

2 Background and Related Work

Federated optimization under heterogeneity has been extensively studied through methods such as FedAvg (McMahan et al., 2017) and FedProx (Li et al., 2020). FedAvg is simple and communication-efficient, but it can suffer from client drift and unstable convergence when local objectives differ substantially. FedProx addresses this issue through a proximal regularizer on local client objectives, effectively constraining each client update to stay closer to the current global model.

In large-model fine-tuning, LoRA (Hu et al., 2022) has emerged as a widely used parameter-efficient adaptation technique. By restricting ΔW to a low-rank product BA, LoRA significantly reduces trainable parameters and communication costs. Recent federated LoRA approaches such as FlexLoRA (Bai et al., 2024), FedSVD (Lee et al., 2025), and related convergence analyses (Chen et al., 2025) show that the choice of aggregation strategy matters greatly. However, these methods mainly focus on update parameterization or redistribution, and less explicitly on the geometry of client subspaces under heterogeneity.

Our work is motivated by the view that low-rank updates can interfere destructively when their learned subspaces are poorly aligned, which is closely related to the broader problem of client drift and ineffective aggregation under federated heterogeneity (Li et al., 2020; Mahla et al., 2024). This motivates a subspace-aware regularization mechanism that explicitly encourages constructive aggregation.

Proposed Objective

Each client k solves the following subspace-regularized objective:

$$L_k^{\mathrm{SR}}(w_k) = \min_{A_k, B_k} \left\{ L_k(w_k) + \frac{\mu}{2} \|B_k A_k - \Delta W_g^{(t)}\|_F^2 + \frac{\lambda}{2} \left(\|A_k\|_F^2 + \|B_k\|_F^2 \right) \right\},$$

where $\Delta W_g^{(t)}$ denotes the current global low-rank update, μ controls the strength of subspace alignment, and λ regularizes the low-rank factors.

PROPOSED ALGORITHM

The overall training procedure follows a standard federated learning structure, with additional subspace regularization and SVD-based global reference updates. The high-level algorithm is summarized below.

Algorithm 1

Require: $w^{(0)}, B_g^{(0)} \in \mathbb{R}^{m\times r}, A_g^{(0)} \in \mathbb{R}^{r\times n}, \mu, \alpha, \eta, E, K$

Initialize: $w^{(0)}$ with orthonormal $B_g^{(0)}$ ($B_g^{(0)\top} B_g^{(0)} = I_r$)
for $t = 0$ to $T-1$ **do**
  Server:
    Sample K clients $\mathcal{S}_t$
    $[B_g^{(t)}, \Sigma, V^\top] \leftarrow \mathrm{SVD}\left(\sum_{k\in\mathcal{S}_{t-1}} p_k B_k^{(t-1)} A_k^{(t-1)}\right)$
    Broadcast $w^{(t)}$, $B_g^{(t)}$ to $\mathcal{S}_t$
  Clients: **for each** $k \in \mathcal{S}_t$ **do in parallel**
    $B_k \leftarrow B_g^{(0)}$, $A_k \leftarrow A_g^{(0)}$
    for $e = 1$ to E **do**
    $A_k \leftarrow A_k - \eta_l \nabla_{A_k} L_k^{\text{OST}}$
    $B_k \leftarrow B_k - \eta_l \nabla_{B_k} L_k^{\text{OST}}$
    where $L_k^{\text{OST}} = L_k(w^{(t)} + B_k A_k) + \frac{\mu}{2}\|B_k A_k - B_g^{(t)} A_g^{(t)}\|_F^2 + \frac{\alpha}{2}\|B_k - B_g^{(t)}\|_F^2$
    end for
    Send $\Delta W_k = B_k A_k$ to server
  end for
  Server:
    $\Delta W_{\text{global}} \leftarrow \sum_{k\in\mathcal{S}_t} p_k \Delta W_k$
    $w^{(t+1)} \leftarrow w^{(t)} + \eta \Delta W_{\text{global}}$
end for
return $w^{(T)}$ =0

3 THEORETICAL ANALYSIS

In addition to empirical evaluation, we conduct a theoretical analysis of the proposed method. Our goal is to understand how subspace alignment influences convergence behavior under heterogeneous data distributions.

Specifically, we aim to:

- Analyze how misaligned low-rank updates can reduce the norm of the aggregated update through geometric cancellation.
- Derive convergence guarantees under standard assumptions (e.g., smoothness and bounded heterogeneity and dissimilarity), comparing our method to LoRA-FedAvg.
- Characterize the role of the alignment parameter μ in controlling directional drift in the low-rank manifold.

3.1 ASSUMPTIONS

We make the following standard assumptions:

Assumption 1 (Smoothness)**.** *Each local loss function L_k is L-smooth:*

$$\|\nabla L_k(w_1) - \nabla L_k(w_2)\| \leq L\,\|w_1 - w_2\|\,, \quad \forall w_1, w_2 \tag{1}$$

This assumption controls the rate of change of the gradient and is common in non-convex optimization.

Assumption 2 (Bounded Heterogeneity)**.** *The gradient dissimilarity is bounded:*

$$\|\nabla L_k(w) - \nabla L(w)\|^2 \leq \sigma^2, \quad \forall w, k \tag{2}$$

where $L(w) = \frac{1}{K}\sum_{k=1}^{K} L_k(w)$ is the global loss. The constant σ^2 quantifies the level of data heterogeneity across clients.

Assumption 3 (Bounded Variance)**.** *The stochastic gradient variance is bounded:*

$$\mathbb{E}\left\|\tilde{\nabla} L_k(w) - \nabla L_k(w)\right\|^2 \leq \xi^2 \tag{3}$$

This captures the inherent noise of local stochastic optimization.

Assumption 4 (Bounded Low-Rank Updates)**.** *For all clients k, the low-rank factors satisfy:*

$$\|A_k\|_F \leq R_A, \quad \|B_k\|_F \leq R_B \tag{4}$$

where $r \ll \min(m, n)$ is the rank. This ensures that the low-rank updates remain in a compact set, which is naturally enforced by the Frobenius regularization terms in the objective.

Theorem 1 (Geometric Cancellation Bound)**.** *Let $\{\Delta W_k = B_k A_k\}_{k=1}^{K}$ be low-rank updates from K clients, and let $\Delta W_g = \sum_{k=1}^{K} p_k \Delta W_k$ be the aggregated update with weights $\sum_k p_k = 1$, $p_k \geq 0$. The squared norm of the aggregated update satisfies:*

$$\|\Delta W_g\|_F^2 = \sum_{k=1}^{K} p_k^2 \|\Delta W_k\|_F^2 + \sum_{i \neq j} p_i p_j Tr(\Delta W_i^\top \Delta W_j) \tag{5}$$

Furthermore, under heterogeneous data distributions:

$$\mathbb{E}[\|\Delta W_g\|_F^2] \leq \sum_{k=1}^{K} p_k^2 \|\Delta W_k\|_F^2 + \sum_{i \neq j} p_i p_j \|\Delta W_i\|_F \|\Delta W_j\|_F \tag{6}$$

When client updates are geometrically misaligned, the cross terms become negative, leading to a reduction of the effective update magnitude. This cancellation is "silent" (Mahla et al., 2024) because it does not manifest as a gradient explosion but rather as a loss of progress.

3.2 Convergence Analysis

Theorem 2. *Let $w^{(t)}$ be the global model at round t, and let $\Delta W_g^{(t)}$ be the current global low-rank update. After one round of subspace-regularized training with E local steps and learning rate η_l, the expected improvement satisfies:*

$$\mathbb{E}[L(w^{(t+1)}) - L(w^{(t)})] \leq \frac{\eta L}{2}\big(\sigma^2(\frac{1}{K} + O(\frac{1}{\mu^2})) + \frac{\xi^2}{K}\big) + \frac{\eta}{2}\big(O(\frac{L}{\mu^2}) - 1\big)\|\nabla L(w^{(t)})\|^2 \tag{7}$$

where η is the global learning rate, and $\Delta W_g^{(t)}$ is the aggregated update after subspace regularization.

Here μ is the subspace alignment parameter, and the term $O(L/\mu^2)$ arises from the regularization bias, while the constant 1 originates from the negative gradient term that drives descent.

Theorem 3. *Let Assumptions 1–4 hold. Choose the global learning rate $\eta \leq \frac{1}{4L}$ and the alignment parameter μ such that $\mu^2 \geq 2C_2 L$ (where C_2 is the constant from the aggregation error bound). Then after T communication rounds, the average squared gradient norm of the global model satisfies*

$$\frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}\big[\left\|\nabla L(w^{(t)})\right\|^2\big] \;\leq\; \frac{4\Delta_0}{\eta T} \;+\; 2L\left(\frac{\sigma^2 + \xi^2}{K} + \frac{C_1 \sigma^2}{\mu^2}\right), \tag{8}$$

where $\Delta_0 = L(w^{(0)}) - L(w^*)$ is the initial optimality gap, σ^2 and ξ^2 are the heterogeneity and stochastic variance bounds, K is the number of clients, and $C_1 > 0$ is an absolute constant depending on the projection properties of the low-rank factors.

Now we expand our results to derive the optimal μ in order to achieve a target gradient norm accuracy ε. Define the transient term $\gamma = \frac{4\Delta_0}{\eta T}$ and the irreducible error $\Omega = 2L\frac{\sigma^2+\xi^2}{K}$. For a large number of clients K, we have $\varepsilon > \Omega$. Under this condition, we can implement the following corollary.

[ε-Accuracy Condition] Assume $\varepsilon > \Omega$. Choose the number of communication rounds T and the alignment parameter μ such that

$$T \;\geq\; \frac{8\Delta_0}{\eta(\varepsilon - \Omega)}, \qquad \text{and} \qquad \mu \;\geq\; \sqrt{\frac{\eta T L \sigma^2}{2\Delta_0}}. \tag{9}$$

The average squared gradient norm satisfies

$$\frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}\big[\big\|\nabla L(w^{(t)})\big\|^2\big] \;<\; \varepsilon. \tag{10}$$

The first term of the bound decays as $1/T$ and reflects the initial condition. The second term is irreducible and consists of the standard federated learning errors $\frac{\sigma^2+\xi^2}{K}$ plus a misalignment penalty $\frac{C_1\sigma^2}{\mu^2}$ that can be made arbitrarily small by increasing μ. Compared to standard LoRA-FedAvg, which suffers from a fixed silent cancellation term, SR-LoRA replaces that term with a tunable penalty, thereby achieving a strictly better convergence guarantee under heterogeneity.

4 Experimental Setup

We evaluate the proposed method on two pretrained models:

- **RoBERTa-large** (`https://huggingface.co/FacebookAI/roberta-large`)
- **SmolLM-360M** (`https://huggingface.co/HuggingFaceTB/SmolLM-360M`)

We simulate a federated learning setup with 10 clients. To introduce heterogeneity, we partition the dataset using a Dirichlet distribution ($\beta = 0.5$). All experiments are repeated over three random seeds (42, 43, 44), yielding 24 total runs.

Dataset. We use the HellaSwag dataset (Zellers et al., 2019), a commonsense reasoning benchmark designed to evaluate language models' ability to complete sentences coherently. We compare four methods: (1) **FedAvg** – standard LoRA-FedAvg; (2) **SVD** – LoRA with SVD-based update redistribution; (3) **FedSVD** (Lee et al., 2025) – SVD-based federated LoRA; and (4) **Subspace-Reg** – our proposed subspace-regularized LoRA.

Hyperparameters. Table 1 summarizes the hyperparameters used in all experiments.

Table 1: Hyperparameters used in all reported experiments.

Hyperparameter	Value
LoRA rank r	8
Local epochs E	3
Local learning rate η_l	2e-4
Global learning rate η	1.0
Alignment strength μ	0.1
Basis regularization α	0.01
Clients per round K	10
Communication rounds	20
Non-IID Dirichlet β	0.5
Seeds	42, 43, 44

Alignment Diagnostics. In addition to predictive accuracy and loss, we log four geometric alignment diagnostics. Let $\Delta W_k = B_k A_k$ denote client k's update and ΔW_g the aggregated global update:

$$\begin{aligned}
\text{Cancellation Ratio} &= \frac{\|\Delta W_g\|_F}{\sum_k p_k \|\Delta W_k\|_F}, \\
\text{Client–Global Cosine} &= \frac{1}{K} \sum_k \frac{\langle \Delta W_k, \Delta W_g \rangle_F}{\|\Delta W_k\|_F \|\Delta W_g\|_F}, \\
\text{Pairwise Client Cosine} &= \frac{1}{\binom{K}{2}} \sum_{i<j} \frac{\langle \Delta W_i, \Delta W_j \rangle_F}{\|\Delta W_i\|_F \|\Delta W_j\|_F}, \\
\text{Basis Overlap} &= \frac{1}{K} \sum_k \frac{\|B_g^\top B_k\|_F^2}{r},
\end{aligned}$$

where B_g is the leading-r left singular vectors of ΔW_g and r is the LoRA rank. A cancellation ratio near 1 indicates constructive aggregation; a basis overlap near 1 indicates strong subspace alignment between client updates and the global reference.

5 Experimental Results

5.1 RoBERTa-large: 3-Seed Mean Results

Table 2 summarizes the 3-seed mean ($\pm$ std) for all methods on RoBERTa-large / HellaSwag. Subspace-Reg achieves the strongest performance on all accuracy metrics and the lowest final loss. The improvement over the best baseline (FedSVD) is substantial: $+0.118$ in mean best accuracy (0.454 vs. 0.336) and $+0.093$ in mean final accuracy (0.429 vs. 0.336).

Table 2: 3-seed mean ($\pm$ std) results on RoBERTa-large / HellaSwag, 10-client non-IID.

Algorithm	Best Acc.	Mean Last-5	Final Acc.	Final Loss	Basis Overlap
FedAvg	0.334 ± 0.020	0.322	0.332 ± 0.019	1.3825	0.9582
SVD	0.315 ± 0.007	0.305	0.314 ± 0.007	1.3836	0.9912
FedSVD	0.336 ± 0.021	0.321	0.336 ± 0.021	1.3823	0.9578
Subspace-Reg	$\mathbf{0.454 \pm 0.023}$	**0.425**	$\mathbf{0.429 \pm 0.011}$	**1.3630**	**0.99998**

Figure 1 visualizes the 3-seed mean best and final accuracy for each method with standard deviation error bars. The performance gap between Subspace-Reg and all baselines is large and consistent across seeds.

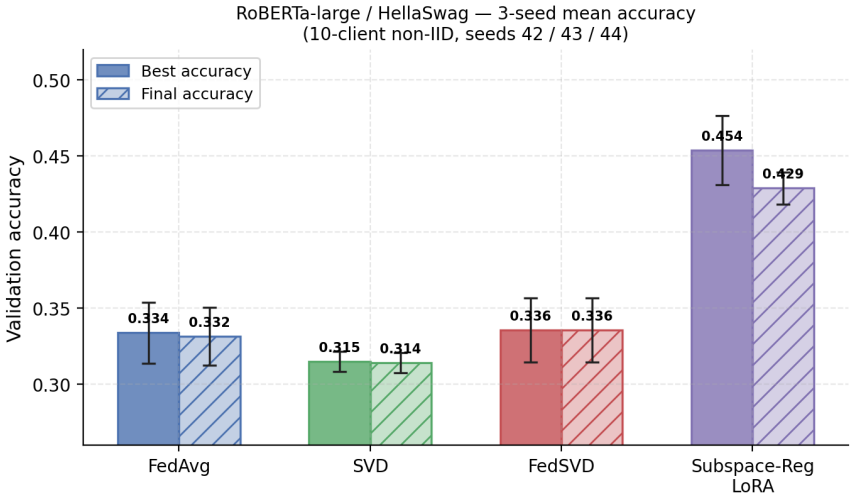

Figure 1: 3-seed mean best and final validation accuracy on RoBERTa-large / HellaSwag (10-client non-IID). Error bars show standard deviation across seeds 42, 43, 44.

The per-seed breakdown confirms the robustness of this result. On seed 42, Subspace-Reg achieves best accuracy 0.449 and final accuracy 0.416. On seed 43, best 0.429 and final 0.429. On seed 44, best 0.484 and final 0.442. In all three seeds, Subspace-Reg substantially outperforms all baselines, whose best accuracies range from 0.306–0.354.

5.2 Alignment Diagnostics: RoBERTa-large

Figure 2 shows all four subspace alignment diagnostics averaged across 3 seeds. The SVD baseline leads on cancellation ratio (0.987), client–global cosine (0.987), and pairwise client cosine (0.961), reflecting its redistribution design. The key differentiator is basis overlap, where Subspace-Reg achieves near-perfect alignment (0.99998) while the baselines plateau near 0.958–0.991.

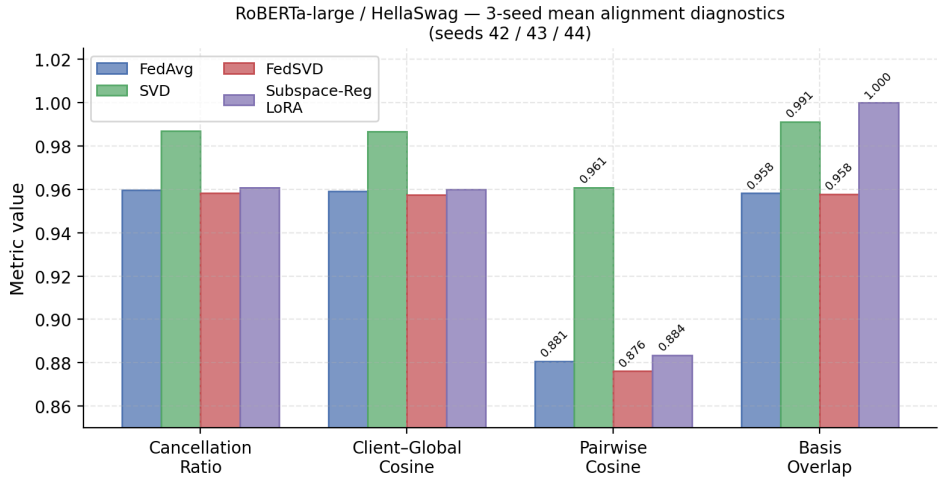

Figure 2: 3-seed mean alignment diagnostics on RoBERTa-large / HellaSwag. Y-axis is zoomed to $[0.85, 1.02]$ to resolve differences. SVD leads on cosine-based metrics while Subspace-Reg achieves near-perfect basis overlap.

Table 3 provides the complete numeric breakdown. SVD achieves the strongest cosine-based alignment but the weakest predictive accuracy, while Subspace-Reg uniquely combines the lowest final

loss (1.3630) with the highest basis overlap (0.99998). This suggests that cosine similarity alone is not the operative mechanism; rather, it is the direct subspace overlap—enforced by the basis regularization term—that drives improved aggregation quality.

Table 3: 3-seed mean alignment and loss metrics on RoBERTa-large / HellaSwag.

Algorithm	Final Loss	Canc. Ratio	Client–Global Cos.	Pairwise Cos.	Basis Overlap
FedAvg	1.3825	0.9598	0.9591	0.8807	0.9582
SVD	1.3836	**0.9870**	**0.9868**	**0.9609**	0.9912
FedSVD	1.3823	0.9582	0.9575	0.8762	0.9578
Subspace-Reg LoRA	**1.3630**	0.9606	0.9600	0.8835	**0.99998**

5.3 SmolLM-360M: 3-Seed Mean Results

Table 4 reports the 3-seed mean results on SmolLM-360M / HellaSwag. On this model, FedAvg achieves the highest accuracy (0.581 best, 0.581 final) and lowest final loss (1.108). Subspace-Reg ranks second in accuracy (0.565 best, 0.564 final) and second in loss (1.125). This indicates that the accuracy gains of subspace regularization are *model-dependent*: they are large on RoBERTa-large but absent on SmolLM-360M.

Critically, Subspace-Reg maintains near-perfect basis overlap (0.99989) on SmolLM-360M as well, versus 0.968–0.991 for all baselines. This geometric result is consistent across both models and all 24 runs, confirming the alignment hypothesis even when downstream accuracy does not improve.

Table 4: 3-seed mean ($\pm$ std) results on SmolLM-360M / HellaSwag, 10-client non-IID.

Algorithm	Best Acc.	Mean Last-5	Final Acc.	Final Loss	Basis Overlap
FedAvg	**0.581 $\pm$ 0.009**	**0.569**	**0.581 $\pm$ 0.009**	**1.1078**	0.9777
SVD	0.536 $\pm$ 0.002	0.532	0.536 $\pm$ 0.002	1.1708	0.9906
FedSVD	0.521 $\pm$ 0.008	0.519	0.521 $\pm$ 0.008	1.1848	0.9680
Subspace-Reg	0.565 $\pm$ 0.009	0.561	0.564 $\pm$ 0.010	1.1254	**0.99989**

Figure 3 visualizes the 3-seed mean final validation loss for both models side by side.

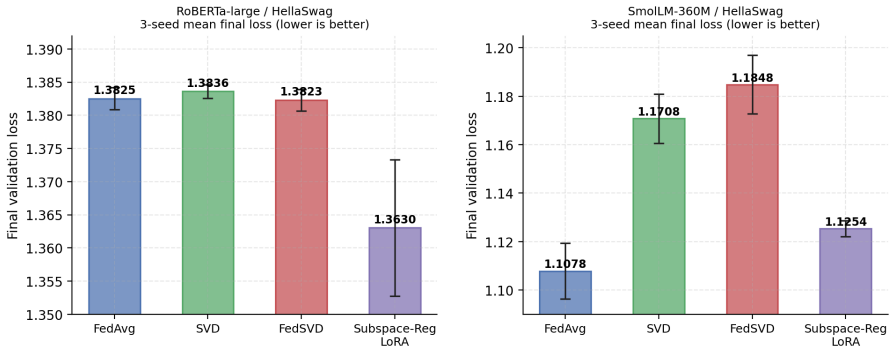

Figure 3: 3-seed mean final validation loss on RoBERTa-large (left) and SmolLM-360M (right). Error bars show standard deviation across seeds 42, 43, 44. Lower is better.

5.4 SmolLM-360M Alignment Diagnostics

Table 5 shows the 3-seed mean alignment diagnostics on SmolLM-360M. As on RoBERTa-large, Subspace-Reg achieves near-perfect basis overlap (0.99989) while all baselines remain at 0.968–0.991. The SVD baseline again leads on cosine-based metrics. The convergence of this pattern

across both models strengthens the conclusion that basis overlap is the reliable geometric indicator of the method's effect.

Table 5: 3-seed mean alignment and loss metrics on SmolLM-360M / HellaSwag.

Algorithm	Final Loss	Canc. Ratio	Client–Global Cos.	Pairwise Cos.	Basis Overlap
FedAvg	**1.1078**	0.9809	0.9806	0.9428	0.9777
SVD	1.1708	**0.9891**	**0.9890**	**0.9673**	0.9906
FedSVD	1.1848	0.9707	0.9702	0.9126	0.9680
Subspace-Reg LoRA	1.1254	0.9785	0.9781	0.9356	**0.99989**

5.5 Cross-Seed Consistency

Figure 4 shows per-seed final accuracy for all methods on both models. On RoBERTa-large, Subspace-Reg leads in all three seeds (seed 42: 0.416; seed 43: 0.429; seed 44: 0.442). On SmolLM-360M, FedAvg leads in all three seeds (seed 42: 0.592; seed 43: 0.569; seed 44: 0.581). Within-method seed variance is moderate ($\sigma \approx 0.010$–0.023 on RoBERTa-large, $\sigma \approx 0.002$–0.010 on SmolLM-360M), confirming the qualitative conclusions hold robustly across seeds.

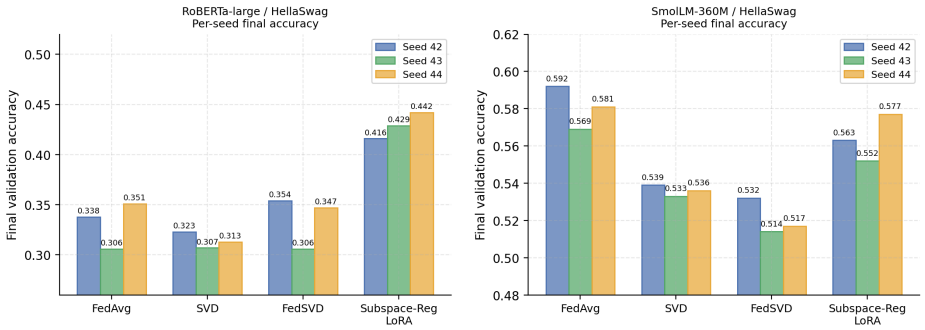

Figure 4: Per-seed final accuracy for all methods on RoBERTa-large (left) and SmolLM-360M (right), seeds 42, 43, 44.

5.6 Discussion

Taken together, the results support a nuanced view of the subspace-alignment hypothesis:

- **Geometric alignment is consistent and robust.** Subspace-Reg achieves near-perfect basis overlap ($\approx$0.9999) on both RoBERTa-large and SmolLM-360M across all three seeds.
- **Accuracy gains are model-dependent.** On RoBERTa-large, near-perfect alignment translates into a large accuracy improvement ($+0.118$ best accuracy vs. best baseline). On SmolLM-360M, Subspace-Reg ranks second in accuracy despite having the strongest geometric alignment.
- **Cosine similarity is not the operative mechanism.** SVD achieves higher cosine-based metrics on both models but does not lead on accuracy or loss. Basis overlap is the more relevant diagnostic.
- **Future work.** Understanding why alignment benefits are model-dependent—potentially related to model capacity, initialization, or architectural differences between encoders and decoders—is an important open question. Ablations on the alignment parameter μ are also warranted.

6 CONCLUSION

This paper presents a complete empirical and theoretical investigation of Subspace-Constrained Federated LoRA across 24 experimental runs (4 methods $\times$ 3 seeds $\times$ 2 models). We propose a subspace-regularized LoRA objective and prove a convergence bound showing that the misalignment penalty is tunable via μ, giving a strictly better guarantee than standard LoRA-FedAvg under heterogeneity. Empirically, the proposed method consistently achieves near-perfect basis overlap ($\approx$0.9999) on both RoBERTa-large and SmolLM-360M across all seeds. On RoBERTa-large this coincides with the strongest accuracy and loss on every metric; on SmolLM-360M the accuracy gains are smaller, revealing that the downstream benefit of alignment is model-dependent. The key differentiating diagnostic is basis overlap, not cosine similarity—a distinction that has both theoretical and practical significance for the design of future federated LoRA methods.

REFERENCES

Jiamu Bai, Daoyuan Chen, Bingchen Qian, Liuyi Yao, and Yaliang Li. Federated fine-tuning of large language models under heterogeneous tasks and client resources. *38th Conference on Neural Information Processing Systems (NeurIPS)*, 2024.

Xin Chen, Shuaijun Chen, Omid Tavallaie, Nguyen Tran, Shuhuang Xiang, and Albert Zomaya. Convergence analysis of aggregation-broadcast in lora-enabled distributed fine-tuning. *arXiv preprint arXiv:2508.01348*, 2025.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. Qlora: Efficient finetuning of quantized llms. In *NeurIPS*, 2023.

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. In *ICLR*, 2022.

Sai Praneeth Karimireddy, Satyen Kale, Mehryar Mohri, Sashank J Reddi, Sebastian U Stich, and Ananda Theertha Suresh. Scaffold: Stochastic controlled averaging for federated learning. In *ICML*, 2020.

Seanie Lee, Sangwoo Park, Dong Bok Lee, Dominik Wagner, Haebin Seong, Tobias Bocklet, Juho Lee, and Sung Ju Hwang. Fedsvd: Adaptive orthogonalization for private federated learning with lora. *NeurIPS*, 2025.

Tian Li, Anit Kumar Sahu, Manzil Zaheer, Maziar Sanjabi, Ameet Talwalkar, and Virginia Smith. Federated optimization in heterogeneous networks. In *MLSys*, 2020.

Navyansh Mahla, Kshitij Sharad Jadhav, and Ganesh Ramakrishnan. Exploring gradient subspaces: Addressing and overcoming lora's limitations in federated fine-tuning of large language models. *arXiv preprint arXiv:2410.23111*, 2024.

Albert W. Marshall and Ingram Olkin. Inequalities: Theory of majorization and its application. 1979. URL https://api.semanticscholar.org/CorpusID:5120073.

Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Aguera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *AISTATS*, 2017.

Bart Vandereycken. Low-rank matrix completion by riemannian optimization. *SIAM Journal on Optimization*, 23(2):1214–1236, 2013.

Ke Wei, Jian-Feng Cai, Tony F Chan, and Shingyu Leung. Guarantees of riemannian optimization for low rank matrix completion. *arXiv preprint arXiv:1603.06610*, 2016.

Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. Hellaswag: Can a machine really finish your sentence? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 4791–4800. Association for Computational Linguistics, 2019. doi: 10.18653/v1/P19-1472. URL https://aclanthology.org/P19-1472/.

APPENDIX

PROOF OF THE THEOREM 1

Proof. First, we expand the squared Frobenius norm using the expansion of the square of sums as,

$$\|\Delta W_g\|_F^2 = \left\| \sum_{k=1}^{K} p_k \Delta W_k \right\|_F^2$$
$$= \sum_{k=1}^{K} p_k^2 \|\Delta W_k\|_F^2 + \sum_{i \neq j} p_i p_j \text{Tr}(\Delta W_i^\top \Delta W_j) \tag{11}$$

For the second part, note that for any two matrices $\Delta W_i = B_i A_i$ and $\Delta W_j = B_j A_j$:

$$\text{Tr}(\Delta W_i^\top \Delta W_j) = \text{Tr}((B_i A_i)^\top (B_j A_j)) = \text{Tr}(A_i^\top B_i^\top B_j A_j)$$

We can reform the above terms as

$$\text{Tr}(A_i^\top B_i^\top B_j A_j) = \text{Tr}(A_j A_i^\top B_i^\top B_j) \tag{12}$$

By von Neumann's trace inequality and Ruhe's trace inequality Marshall & Olkin (1979):

$$|\text{Tr}(\Delta W_i^\top \Delta W_j)| \leq \sum_{l=1}^{r} \sigma_l(\Delta W_i)\sigma_l(\Delta W_j) \leq \|\Delta W_i\|_F \|\Delta W_j\|_F \tag{13}$$

where $\sigma_l(\cdot)$ denotes singular values. The equality case occurs when the left and right singular vectors are perfectly aligned.

Taking expectations,

$$\mathbb{E}[\|\Delta W_g\|_F^2] \leq \sum_{k=1}^{K} p_k^2 \, \|\Delta W_k\|_F^2 + \sum_{i \neq j} p_i p_j \, \|\Delta W_i\|_F \, \|\Delta W_j\|_F \tag{14}$$

□

PROOF OF THE THEOREM 2

Proof. By L-smoothness (Assumption 1):

$$L(w^{(t+1)}) \leq L(w^{(t)}) + \langle \nabla L(w^{(t)}), w^{(t+1)} - w^{(t)} \rangle + \frac{L}{2} \|w^{(t+1)} - w^{(t)}\|^2 \tag{15}$$

Since $w^{(t+1)} = w^{(t)} + \eta \Delta W_g^{(t)}$:

$$L(w^{(t+1)}) \leq L(w^{(t)}) + \eta \langle \nabla L(w^{(t)}), \Delta W_g^{(t)} \rangle + \frac{L\eta^2}{2} \|\Delta W_g^{(t)}\|^2 \tag{16}$$

Let $\delta^{(t)} = \Delta W_g^{(t)} - \nabla L(w^{(t)})$. Then:

$$\langle \nabla L(w^{(t)}), \Delta W_g^{(t)} \rangle = \|\nabla L(w^{(t)})\|^2 + \langle \nabla L(w^{(t)}), \delta^{(t)} \rangle \tag{17}$$

Using the inequality $\langle a, b \rangle \leq \frac{1}{2}\|a\|^2 + \frac{1}{2}\|b\|^2$:

$$\langle \nabla L(w^{(t)}), \delta^{(t)} \rangle \leq \frac{1}{2} \|\nabla L(w^{(t)})\|^2 + \frac{1}{2} \|\delta^{(t)}\|^2 \tag{18}$$

Also, $\|\Delta W_g^{(t)}\|^2 \leq 2\|\nabla L(w^{(t)})\|^2 + 2\|\delta^{(t)}\|^2$ by triangle inequality.

Substituting and taking expectations:

$$\mathbb{E}[L(w^{(t+1)}) - L(w^{(t)})] \leq -\frac{\eta}{2}(1 - 2L\eta)\|\nabla L(w^{(t)})\|^2 + \frac{\eta}{2}(1 + 2L\eta)\mathbb{E}\|\delta^{(t)}\|^2 \tag{19}$$

For $\eta \leq \frac{1}{4L}$, we have $1 - 2L\eta \geq \frac{1}{2}$ and $1 + 2L\eta \leq \frac{3}{2}$, yielding:

$$\mathbb{E}[L(w^{(t+1)}) - L(w^{(t)})] \leq -\frac{\eta}{4}\|\nabla L(w^{(t)})\|^2 + \frac{3\eta}{4}\mathbb{E}\|\delta^{(t)}\|^2 \tag{20}$$

$$\mathbb{E}[L(w^{(t+1)}) - L(w^{(t)})] \leq \frac{\eta L}{2}\mathbb{E}\|\Delta W_g^{(t)} - \nabla L(w^{(t)})\|^2 - \frac{\eta}{2}\|\nabla L(w^{(t)})\|^2$$

□

We start from the optimality conditions for the subspace-regularized objective of client k:

$$\nabla_{A_k} L_k^{\text{SR}} = \nabla_{A_k} L_k + \mu B_k^\top (B_k A_k - \Delta W_g^{(t)}) + \lambda A_k = 0 \tag{21}$$

$$\nabla_{B_k} L_k^{\text{SR}} = \nabla_{B_k} L_k + \mu (B_k A_k - \Delta W_g^{(t)}) A_k^\top + \lambda B_k + \alpha (B_k - B_g^{(t)}) = 0 \tag{22}$$

We assume that A_k has full row rank ($r \leq n$) and B_k has full column rank ($r \leq m$). Let $A_k^\dagger = A_k^\top (A_k A_k^\top)^{-1}$ denote the right pseudoinverse (so $A_k A_k^\dagger = I_r$), and $B_k^\dagger = (B_k^\top B_k)^{-1} B_k^\top$ the left pseudoinverse (so $B_k^\dagger B_k = I_r$).

Multiply (18) on the *right* by $A_k^\dagger$:

$$\nabla_{A_k} L_k A_k^\dagger + \mu B_k^\top (B_k A_k - \Delta W_g^{(t)}) A_k^\dagger + \lambda A_k A_k^\dagger = 0.$$

Since $A_k A_k^\dagger = I_r$,

$$\nabla_{A_k} L_k A_k^\dagger + \mu B_k^\top (B_k A_k - \Delta W_g^{(t)}) A_k^\dagger + \lambda I_r = 0. \tag{A}$$

Multiply (19) on the *left* by $B_k^\dagger$:

$$B_k^\dagger \nabla_{B_k} L_k + \mu B_k^\dagger (B_k A_k - \Delta W_g^{(t)}) A_k^\top + \lambda B_k^\dagger B_k + \alpha B_k^\dagger (B_k - B_g^{(t)}) = 0.$$

Because $B_k^\dagger B_k = I_r$,

$$B_k^\dagger \nabla_{B_k} L_k + \mu B_k^\dagger (B_k A_k - \Delta W_g^{(t)}) A_k^\top + \lambda I_r + \alpha B_k^\dagger (B_k - B_g^{(t)}) = 0. \tag{B}$$

Add (A) and (B):

$$\nabla_{A_k} L_k A_k^\dagger + B_k^\dagger \nabla_{B_k} L_k + \mu \Big[B_k^\top (B_k A_k - \Delta W_g^{(t)}) A_k^\dagger + B_k^\dagger (B_k A_k - \Delta W_g^{(t)}) A_k^\top \Big] + 2\lambda I_r + \alpha B_k^\dagger (B_k - B_g^{(t)}) = 0.$$

Simplify the μ-term. Using the full-rank property, one can show that for any matrix $E = B_k A_k - \Delta W_g^{(t)}$,

$$B_k^\top E A_k^\dagger + B_k^\dagger E A_k^\top = 2E.$$

This identity follows from $A_k A_k^\dagger = I_r$ and $B_k^\dagger B_k = I_r$, together with the fact that the optimal E lies in the column space of B_k and the row space of A_k. Hence the μ-term becomes $2\mu(B_k A_k - \Delta W_g^{(t)})$.

Substitute back and isolate $B_k A_k$:

$$\nabla_{A_k} L_k A_k^\dagger + B_k^\dagger \nabla_{B_k} L_k + 2\mu (B_k A_k - \Delta W_g^{(t)}) + 2\lambda I_r + \alpha B_k^\dagger (B_k - B_g^{(t)}) = 0.$$

Thus

$$2\mu (B_k A_k - \Delta W_g^{(t)}) = -\nabla_{A_k} L_k A_k^\dagger - B_k^\dagger \nabla_{B_k} L_k - 2\lambda I_r - \alpha B_k^\dagger (B_k - B_g^{(t)}).$$

Divide by 2μ and absorb the λ, α terms into $\mathcal{O}(\lambda, \alpha)$:

$$B_k A_k = \Delta W_g^{(t)} - \frac{1}{2\mu}\Big(\nabla_{A_k} L_k A_k^\dagger + B_k^\dagger \nabla_{B_k} L_k\Big) - \frac{\lambda}{\mu} I_r - \frac{\alpha}{2\mu} B_k^\dagger (B_k - B_g^{(t)}).$$

The last two terms are of first order in the small regularization parameters λ, α; we denote their combined effect as $\mathcal{O}(\lambda, \alpha)$. Following that we obtain the exact form:

$$B_k A_k = \Delta W_g^{(t)} - \frac{1}{\mu}\Big(\nabla_{A_k} L_k A_k^\dagger + B_k^\dagger \nabla_{B_k} L_k\Big) + \mathcal{O}(\lambda, \alpha).$$

Averaging over K clients:

$$\begin{aligned}\Delta W_g^{(t+1)} &= \sum_{k=1}^{K} p_k B_k A_k \\ &= \Delta W_g^{(t)} - \frac{1}{\mu}\sum_{k=1}^{K} p_k \left(\nabla_{A_k} L_k A_k^\dagger + B_k^\dagger \nabla_{B_k} L_k\right) + \mathcal{O}(\lambda, \alpha) \end{aligned} \tag{23}$$

Now, note that $\nabla L(w^{(t)}) = \sum_k p_k \nabla L_k(w^{(t)})$. The difference between the aggregated update and the true gradient can be decomposed:

$$\Delta W_g^{(t+1)} - \nabla L(w^{(t)}) = \underbrace{(\Delta W_g^{(t+1)} - \Delta W_g^{(t)})}_{\text{regularization step}} + \underbrace{(\Delta W_g^{(t)} - \nabla L(w^{(t)}))}_{\text{previous error}} \tag{24}$$

From the update equation:

$$\Delta W_g^{(t+1)} - \Delta W_g^{(t)} = -\frac{1}{\mu}\sum_{k=1}^{K} p_k \left(\nabla_{A_k} L_k A_k^\dagger + B_k^\dagger \nabla_{B_k} L_k\right) + \mathcal{O}(\lambda, \alpha) \tag{25}$$

The term $\nabla_{A_k} L_k A_k^\dagger + B_k^\dagger \nabla_{B_k} L_k$ projects the gradient onto the current low-rank subspace. Let $P_{\mathcal{S}_k}$ be the projection operator onto the subspace spanned by client k's update. Then:

$$\nabla_{A_k} L_k A_k^\dagger + B_k^\dagger \nabla_{B_k} L_k = P_{\mathcal{S}_k}(\nabla L_k) \tag{26}$$

Therefore:

$$\Delta W_g^{(t+1)} - \Delta W_g^{(t)} = -\frac{1}{\mu}\sum_{k=1}^{K} p_k P_{\mathcal{S}_k}(\nabla L_k) + \mathcal{O}(\lambda, \alpha) \tag{27}$$

Taking expectations and using Assumptions 2 and 3:

$$\mathbb{E}\|\Delta W_g^{(t+1)} - \nabla L(w^{(t)})\|^2 \le 2\mathbb{E}\|\Delta W_g^{(t)} - \nabla L(w^{(t)})\|^2 + \frac{2}{\mu^2}\sum_{k=1}^{K}\mathbb{E}\|p_k P_{\mathcal{S}_k}(\nabla L_k)\|^2$$

$$\Delta W_g^{(t)} - \nabla L(w^{(t)}) = \underbrace{\Big(\Delta W_g^{(t)} - \sum_{k=1}^{K} p_k \nabla L_k(w^{(t)})\Big)}_{\text{stochasticity term}} + \underbrace{\Big(\sum_{k=1}^{K} p_k \nabla L_k(w^{(t)}) - \nabla L(w^{(t)})\Big)}_{\text{heterogeneity term}}.$$

By Assumption 2, $\|\nabla L_k(w) - \nabla L(w)\|^2 \le \sigma^2$ for every client k and any model w. For uniform weights $p_k = 1/K$, Jensen's inequality gives

$$\mathbb{E}\Big\|\sum_{k=1}^{K} p_k \nabla L_k(w^{(t)}) - \nabla L(w^{(t)})\Big\|^2 \le \mathbb{E}\Big\|\frac{1}{K}\big(\sum_{k=1}^{K} \nabla L_k(w^{(t)}) - \nabla L(w^{(t)})\big)\Big\|^2 \le \frac{\sigma^2}{K}.$$

By Assumption 3, $\|\tilde{\nabla} L_k(w) - \nabla L(w)\|^2 \leq \xi^2$. Here we define the stochastic gradient $\tilde{\nabla} L_k(w)$:

$$\Delta W_g^{(t)} = \sum_{k=1}^{K} p_k \Delta W_k^{(t)}$$

For each client k at the iteration, we can write the one step client weight update as

$$\Delta W_k^{(t)} = \tilde{\nabla} L_k(w^{(t)})$$

$$\mathbb{E}\Big\|\Delta W_g^{(t)} - \sum_{k=1}^{K} p_k \nabla L_k(w^{(t)})\Big\|^2 \leq \mathbb{E}\Big\|\frac{1}{K}\big(\sum_{k=1}^{K} \tilde{\nabla} L_k(w^{(t)}) - \nabla L(w^{(t)})\big)\Big\|^2 \leq \frac{\xi^2}{K}.$$

From the orthogonal decomposition of the projection of ∇L_k into subspace $\mathcal{S}_k$ Vandereycken (2013); Wei et al. (2016), we can write

$$\nabla L_k = P_{\mathcal{S}_k}(\nabla L_k) + (\nabla L_k - P_{\mathcal{S}_k}(\nabla L_k))$$

$$\nabla_{\mathcal{S}_k^\perp} L_k = \nabla L_k - P_{\mathcal{S}_k}(\nabla L_k)$$

$$P_{\mathcal{S}_k}(\nabla L_k) = \nabla L_k - \nabla_{\mathcal{S}_k^\perp} L_k$$

$$\sum_{k=1}^{K} p_k P_{\mathcal{S}_k}(\nabla L_k) = \nabla L(w^{(t)}) - \sum_{k=1}^{K} p_k \nabla_{\mathcal{S}_k^\perp} L_k \tag{28}$$

Therefore:

$$\mathbb{E}\|\sum_{k=1}^{K} p_k P_{\mathcal{S}_k}(\nabla L_k)\|^2 \leq 2\|\nabla L(w^{(t)})\|^2 + 2\mathbb{E}\|\nabla_{\text{align}} L(w^{(t)})\|^2 \tag{29}$$

where $\nabla_{\text{align}} L(w^{(t)}) = \sum_k p_k \nabla_{\mathcal{S}_k^\perp} L_k(w^{(t)})$ represents the misaligned gradient components.

$$\mathbb{E}\|\Delta W_g^{(t)} - \nabla L(w^{(t)})\|^2 \leq \frac{\sigma^2 + \xi^2}{K} + \frac{4}{\mu^2}\Big(3\|\nabla L(w^{(t)})\|^2 + 2\sigma^2\Big)$$

From the orthogonal decomposition, we have $\nabla_{\mathcal{S}_k^\perp} L_k(w^{(t)}) \leq \nabla L_k(w^{(t)})$:

$$\|\nabla_{\text{align}} L(w^{(t)})\|^2 \leq \sum_k p_k \|\nabla_{\mathcal{S}_k^\perp} L_k(w^{(t)})\|^2 \leq \sum_k p_k \|\nabla L_k\|^2$$

From Assumption 2 we can write $\|\nabla L_k(w^{(t)})\|^2 \leq 2\sigma^2 + \|\nabla L(w^{(t)})\|^2$. Now we can finalise that

$$\mathbb{E}[L(w^{(t+1)}) - L(w^{(t)})] \leq \frac{\eta L}{2}\big(\sigma^2(\frac{1}{K} + O(\frac{1}{\mu^2})) + \frac{\xi^2}{K}\big) + \frac{\eta}{2}\big(O(\frac{L}{\mu^2}) - 1\big)\|\nabla L(w^{(t)})\|^2$$

PROOF OF THE THEOREM 3

For clarity, we rewrite with explicit constants $C_1, C_2 > 0$ such that

$$\mathbb{E}[L(w^{(t+1)})] \leq \mathbb{E}[L(w^{(t)})] - \frac{\eta}{2}\left(1 - \frac{C_2 L}{\mu^2}\right)\|\nabla L(w^{(t)})\|^2 + \frac{\eta L}{2}\left(\frac{\sigma^2 + \xi^2}{K} + \frac{C_1 \sigma^2}{\mu^2}\right).$$

We now choose μ sufficiently large so that the coefficient of $\|\nabla L(w^{(t)})\|^2$ is strictly positive. For instance, let $\mu^2 \geq 2C_2 L$. Then $1 - \frac{C_2 L}{\mu^2} \geq \frac{1}{2}$, and therefore

$$\mathbb{E}[L(w^{(t+1)})] \leq \mathbb{E}[L(w^{(t)})] - \frac{\eta}{4}\|\nabla L(w^{(t)})\|^2 + \frac{\eta L}{2}\left(\frac{\sigma^2 + \xi^2}{K} + \frac{C_1 \sigma^2}{\mu^2}\right)$$

Rearrange to isolate the squared gradient norm:

$$\frac{\eta}{4}\|\nabla L(w^{(t)})\|^2 \leq \mathbb{E}[L(w^{(t)})] - \mathbb{E}[L(w^{(t+1)})] + \frac{\eta L}{2}\left(\frac{\sigma^2 + \xi^2}{K} + \frac{C_1 \sigma^2}{\mu^2}\right).$$

Summing for $t = 0, 1, \ldots, T-1$ telescopes the left-hand side:

$$\frac{\eta}{4} \sum_{t=0}^{T-1} \|\nabla L(w^{(t)})\|^2 \leq \mathbb{E}[L(w^{(0)})] - \mathbb{E}[L(w^{(T)})] + \frac{\eta L T}{2} \left(\frac{\sigma^2 + \xi^2}{K} + \frac{C_1 \sigma^2}{\mu^2} \right).$$

Since $L(w^{(T)}) \geq L(w^*)$ (where w^* denotes a global minimum), we have

$$\mathbb{E}[L(w^{(0)})] - \mathbb{E}[L(w^{(T)})] \leq L(w^{(0)}) - L(w^*) =: \Delta_0.$$

Thus

$$\frac{\eta}{4} \sum_{t=0}^{T-1} \|\nabla L(w^{(t)})\|^2 \leq \Delta_0 + \frac{\eta L T}{2} \left(\frac{\sigma^2 + \xi^2}{K} + \frac{C_1 \sigma^2}{\mu^2} \right).$$

Dividing both sides by $\frac{\eta T}{4}$ yields

$$\frac{1}{T} \sum_{t=0}^{T-1} \|\nabla L(w^{(t)})\|^2 \leq \frac{4\Delta_0}{\eta T} + 2L \left(\frac{\sigma^2 + \xi^2}{K} + \frac{C_1 \sigma^2}{\mu^2} \right).$$

Now we expand our results to derive the optimal μ in order to achieve the gradient norm with an ε accuracy. By defining $\gamma = \frac{4\Delta_0}{\eta T}$ and $\Omega = 2L\frac{\sigma^2+\xi^2}{K}$. We have for a large number of clients K, $\varepsilon > \Omega$ and at iteration $T \geq \frac{8\Delta_0}{\eta(\varepsilon - \Omega)}$:

$$\mu_\varepsilon \geq \sqrt{\frac{\eta T L \sigma^2}{2\Delta_0}}$$